# Coupled Mission Planning and Facility Process Modeling for Hanford High-Level Waste Vitrification

*Architecture and algorithms of the EWG campaign builder and the HLW system model*

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## Executive summary

This paper documents two coupled software applications that model the Hanford tank waste retrieval and vitrification mission end to end. The first, the EWG campaign builder, turns tank-by-tank waste inventory data into a sequenced set of retrieval campaigns, each with a feed blend, a waste acceptance criteria (WAC) evaluation, and an optimized glass-forming chemical addition. The second, the HLW system model, takes each campaign blend as speciated feed and simulates the plant that processes it: the melter feed prep vessel, the melter feed vessel, the joule-heated melter, and the complete off-gas train from the submerged bed scrubber to the stack, plus the condensate (effluent) system and the discrete operator-procedure schedule that ties the plant to a calendar.

The two applications share a single data lineage. Best-Basis Inventory (BBI) chemistry, downloaded from the PNNL PHOENIX service (October 2026), is converted to per-layer feed vectors using the layer-by-layer convention of RPP-RPT-64878 Appendix D, extended with homogenized heel layers for every waste tank not covered by a published layer. This includes all tanks in the NE, SW, and NW tank farms to provide a bounding estimate for the total HLW mission duration. A campaign-generation algorithm taken from RPP-RPT-64878 Section 6.2 assembles campaigns under tank-order, quadrant-sequencing, and volume-band rules, then refines each blend with a linear program. Glass formulation for every campaign comes from the EWG2.5 property and composition models (PNNL-35884), solved as a constrained nonlinear program that maximizes waste loading subject to product consistency, viscosity, electrical conductivity, sulfate, phosphate, and composition-validity constraints.

The facility model is built as a chain of unit-operation modules with one shared thermodynamic core. Every stream carries molar flow per chemical species, phase, temperature, and pressure, and every unit closes both mass and energy against that core. The melter itself is an element-ledger partition engine: feed species decompose to elements, scenario-supplied decontamination factors move a computed fraction of each volatile element to the off gas, and the remainder resynthesizes as glass oxides. Downstream stages, a submerged bed scrubber scrubber, wet electrostatic precipitator, mist eliminator, reheater, high-efficiency particulate air filter, activated carbon and silver-mordenite adsorber beds, a regenerative heat exchanger coupled to selective catalytic oxidation and reduction reactors, each apply per-species capture factors and thermodynamic state changes. An equilibrium condensate-chemistry model tracks the transfer schedule and pH of the scrubber liquor on its way to the facility effluent.

In order to handle HLW campaigns built from future retrieval of the NW, NE, and SW quadrants of the tank farm, wash-efficiency control re-synthesizes the heel records at any specified fraction of soluble species decanted to the low-activity waste plant during retrieval. Across two scenarios presented as an example in this white paper, lowering the wash efficiency from 0.9 to 0.5 nearly doubles mission glass (45.0 to 86.2 kt on scenario 135.105.200) and extends mission runtime from 9,044 to 16,033 days. Second, a melter-feed monitoring chain exposes computed composition streams at three points: the blended slurry feed, the melter feed prep vessel outlet, and the melter feed vessel outlet that the melter actually draws. The chain closes to 100.00% on campaign mass, and the prep vessel and feeder vessel outlets agree to within 0.1% molar composition. Third, a master export writes every computed stream of a campaign, with rates and campaign-integrated masses, into a single auditable Excel workbook.

Verification is built into the workflow rather than bolted on. The suite runs 147 automated tests, including per-species mass closure with hybrid tolerances, energy closure with explicit duty reporting, physical anchors such as saturation-point checks, and a reagent-inclusive element audit across the entire train boundary. On a representative 59-campaign mission, all 38 tracked elements close to better than 1% relative, with the single worst offender a trace arsenic flow of about 0.2 milligrams per second whose 21% relative residual is 1 microgram per second in absolute terms. The model runs a full campaign in 1 to 2 seconds and a full mission in about 76 seconds on a desktop workstation, fast enough to support the sensitivity sweeps reported in the Results section.

The Methodology section of this paper covers the architecture and algorithms in depth. The Results section reports mission totals, campaign anatomy, wash sensitivity, off-gas performance, and verification evidence. The Discussion section addresses model limitations and data provenance candidly.

# Introduction

### Mission context

The Hanford tank farms hold roughly 56 million gallons of mixed radioactive and chemical waste in 177 underground tanks, a legacy of plutonium production ending in the 1980s. The mission currently baselined by the U.S. Department of Energy retrieves that waste, separates most of the volume and sodium salts into a low-activity waste (LAW) stream for immobilization, and converts the remainder, the sludge fraction bearing the transuranic and cesium-strontium

activity, into glass in a high-level waste (HLW) vitrification facility. Retrieval sequencing matters because waste is heterogeneous at every scale: between tanks, between horizontal layers within a tank, and between the supernatant liquor, saltcake, and sludge phases within a layer. Every decision about which material moves when, and how much wash water follows it, propagates directly into the size of the HLW mission: how much glass, how many canisters, how many years.

Planning analyses for this mission have historically split the problem at the facility fence line. Upstream studies produce feed schedule and blend composition. Downstream studies assume a feed and size the plant. The coupling is where the interesting engineering lives. A campaign blend that satisfies waste acceptance criteria on paper can still drive the off-gas train hard, because soluble nitrate and sulfate that survive retrieval washing volatilize in the melter and load the scrubber and catalytic reactors. A feed that is comfortable for the melter can strand heel material that doubles the number of campaigns. The two applications described here were built to close that gap with one continuous data path from tank inventory chemistry to stack composition and operator schedules.

## The two applications

The EWG campaign builder is the upstream application. It ingests the Best-Basis Inventory, constructs per-layer feed vectors following the Appendix D layer-by-layer convention of RPP-RPT-64878, synthesizes homogenized heel layers for tanks outside the published layer set, and generates retrieval campaigns under the sequencing rules of that document's Section 6.2. For each campaign it blends layer vectors by volume fraction, evaluates the blend against WAC limits, and solves for the glass-forming chemical addition that maximizes waste loading under the EWG2.5 property models of PNNL-35884. Its native outputs are the campaign recipe (which layers, what fractions), the per-campaign waste oxide vector, the glass formulation and projected glass properties, and mission-level rollups.

The HLW system model is the downstream application. It converts each campaign recipe into a speciated chemical feed, named ions, hydrates, isotopes, and molecular species rather than lumped oxides, and simulates the plant unit by unit. Feed flows from the melter feed prep vessel (MFPV) through the melter feed vessel (MFV) into a joule-heated melter. The melter partitions volatile species into an off-gas stream that passes through a wet train (submerged bed scrubber, wet electrostatic precipitator, mist eliminator), a reheater and high-efficiency particulate air filter stage, a fan, packed adsorber beds for mercury and iodine, and a hot side comprising a regenerative heat exchanger, selective catalytic oxidation, and selective catalytic reduction,

before discharge. Condensate from the wet stages accumulates in a vessel whose transfers are scheduled and whose chemistry is tracked. A discrete operations layer schedules the operator procedures, batch receipt, dosing, transfers, and pours, against the physical model's batch volumes, so the model answers not only how much but when. The discreet modeling layer is a point of future development and only exists in the current model as a ‘proof of concept’.

## Scope of this paper

Section 2 (Methodology) documents the architecture of both applications and the algorithms that produce their results, at the level of detail an engineer needs to reproduce a number or audit an assumption. Section 3 (Results) reports verified mission-level and campaign-level outputs for two published scenarios, the new wash-efficiency sensitivity sweep, off-gas and effluent performance, and the verification evidence. Section 4 (Discussion) covers interpretation, limitations, data provenance caveats, and computational performance. Section 5 states conclusions. All quantitative results in this paper were produced by the version of the code currently on the repository's main branch (as of 12 October 2026), and every headline number was re-derived from cached model outputs at the time of writing.

# Methodology

## System architecture overview

Both applications are Python packages. The EWG campaign builder lives in its own repository and exposes a small set of functions the facility model imports as a library: scenario construction, layer vector loading, campaign recipe generation, and glass formulation. The HLW system model is organized as a unit-operation chain. Each unit operation lives in its own module and implements one common interface: it consumes one or more inlet streams plus a parameter dictionary, and returns outlet streams, one or more duty terms in kilowatts, and a diagnostics dictionary. This contract is what makes the chain extensible, a new wet scrubbing stage, for example, is one YAML parameter block and a driver of a few dozen lines, because the shared wet-stage engine handles capture, saturation, and the energy balance.

Three layers sit below the unit operations. A species library defines every chemical species the model tracks, ideal gases with NASA Shomate polynomial thermodynamics, aqueous ions, condensed oxides and hydrates, radioisotopes, and isotope carrier species. A Stream class carries molar flow per species (mol/s), a phase map, temperature, and pressure, and computes enthalpy as the sum over species of formation enthalpy plus sensible heat, evaluated in the species' own phase. A balances module provides mixers, splitters, and saturation flashes used by many units. Everything above this core is composition-agnostic: a unit never needs to know the full species list, only the capture or reaction behavior of the species it acts on.

A Streamlit application provides the engineer-facing surface: a process flow diagram tab that redraws from live model state, stream composition tables, per-unit energy, effluent chemistry, media loading ledgers, mission-level trend plots fed by a cached sweep of all campaigns, a custom-trend builder over any computed variable, an export tab whose master workbook carries every stream, and an operations tab that renders the discrete-event schedule. Each tab body runs behind an error boundary so a data-path failure in one view does not take down the application.

## Data lineage: from BBI to campaign feed

The upstream data source is the Best-Basis Inventory, retrieved from the PNNL PHOENIX data service (October 2026) as JSON and flattened to a per-tank chemistry snapshot. Retrieval of roughly 400 MB is scripted with a browser-identical user agent, and the ingestion code verifies response shape before parsing, because the service's front-end can return an HTTP 200 page containing a rejection notice instead of data.

Layer vectors come from the Appendix D workbook of RPP-RPT-64878, which publishes, for each published scenario condition (supernatant management, target waste loading, and solids loading combinations), a per-layer feed composition and per-layer waste acceptance parameter values. The builder reads each layer's liquid and solid inventory rows, about 212 analytes per layer spanning elements, ions, hydrates, isotopes, and bound-species bookkeeping keys, and sums them through a shared element-to-oxide stoichiometry table. The stoichiometry uses the n-atoms-per-oxide form, for example the aluminum factor is the molar mass of Al2O3 divided by twice the molar mass of Al, not a bare oxide-to-element molar mass ratio, and an anion list (carbonate, nitrate, nitrite, hydroxide, sulfate, phosphate, oxalate, halides, free water, and bound-oxygen keys) is excluded from oxide mass and routed to the WAP chemistry instead. The anion exclusion matters at the scale of tons: carbonate alone carries roughly 53 t in a single large tank layer, and mapping it naively to a metal oxide by name similarity fabricates about a kilotonne of phantom cobalt oxide across the published layer set.

Tanks without a published layer (all quadrants outside of SE), principally the sludge heel material left after supernatant and saltcake retrieval, are appended as homogenized heel layers synthesized from whole-tank BBI chemistry, one per tank, named with a conventional suffix. Sludge classification uses the waste-type field: only sludge-type wastes feed the HLW path, while saltcake, salt slurry, evaporator product, and liquid waste types route to LAW and are excluded from HLW layer mass. The resulting layer set for the sanitized mission is 38 published layers plus 53 heel layers, about 15 Mt of waste oxide in total.

Campaign recipes follow the Section 6.2 retrieval plan of RPP-RPT-64878. The builder walks tanks and layers under sequencing rules: an in-flight layer continues until empty; a layer is eligible only when it is the shallowest unused layer in its tank; each tank contributes at least a minimum transfer volume; tank-order precedence rules apply within the southeast quadrant; quadrants drain in order (southeast first, then northeast, northwest, southwest); a campaign blends no more than five tanks; and campaign volume stays inside a 450 to 550 kgal band. Within each campaign, a linear program (scipy's HiGHS solver) maximizes waste mass transferred subject to the volume band, per-layer availability, a per-layer take cap that keeps candidate sets from collapsing to one or two tanks, and linearized WAC constraints. Campaign counts for the two scenarios studied here are 59 (condition 135.105.200) and 76 (condition 135.120.150), and the counts are read live from the builder rather than hard-coded in any downstream test, a deliberate choice after a stale-count regression.

The facility model consumes recipes, not raw layer files. Its bridge module takes a campaign's layer list and volume fractions, retrieves each layer's speciated analyte masses, folds them by fraction into a campaign analyte inventory, and converts analytes to named chemical species. The conversion follows rules locked with the subject matter experts: metal-bound analyte keys (sodium-bound, potassium-bound, chloride-bound) are real mass routed to the host cation; bound-oxygen and bound-water keys are bookkeeping and dropped; explicit hydrates such as aluminum hydroxide and oxyhydroxide stay molecular so the plant model can evolve them (boil-off, dehydration) rather than pre-oxidizing them; isotopes are kept as explicit species and pooled onto their host element at the melter ledger; and every species unknown to the library is collected into a missing-species list that is driven to zero, 83 of 83 tracked on the reference campaign, rather than silently dropped.

### Glass formulation: EWG2.5 as a constrained optimization

For each campaign the builder solves for the glass-forming chemical addition that maximizes waste loading. The formulation models of PNNL-35884 are treated as constraint functions. Product consistency (the normalized release *NL* from the product consistency test) is an eleven-term plus residual polynomial quadratic model. Viscosity and electrical conductivity use Vogel-Fulcher-Tammann forms with composition-linear coefficients. Sulfate solubility uses a multi-term quadratic mixture model with a trisodium sulfate monoxide basis and a small negative offset. Phosphate capacity is a logistic model whose published coefficient table has no intercept by design; the sodium-silicate cross-term mediates what the narrative text describes, and mapping prose onto coefficients by hand reproduces neither the paper's example glasses nor its reported predictions. Each model additionally carries per-oxide validity bounds and an immiscibility constraint on the sodium-lithium-borosilicate ratio.

The solve is sequential least squares programming (SLSQP) with two reliability measures. First, the solver is run multi-start over a grid of initial waste fractions; a single start is infeasible-by-convergence on high-sodium wastes, where the solver's line search dies before finding the feasible basin, while low-waste starts are feasible by construction and climb to the optimum. The best feasible result across starts is kept, and feasibility means both solver success and every constraint passing. Second, raw glass-former demands are recovered through per-former oxide yields (for example boric acid carries 0.563 as-fired $B_2O_3$), because the solver's decision variables are oxide mass fractions of the final glass, not powder masses; treating them as powder masses underfeeds boron by roughly 44%.

Model basis conventions matter as much as the coefficients. Each property model evaluates on its own declared basis: the product consistency model uses eleven modeled oxides plus a residual Others term that absorbs all remaining mass; viscosity uses fifteen Vogel-Fulcher-Tammann coefficient components plus its own Others; the sulfate model renormalizes to a sulfate-free basis internally. Evaluating all models on one shared full composition basis biases viscosity low by 12 to 25% and normalized release low by 30 to 70% against the published example glasses. The implementation reproduces the published Table 3-6 property values for the paper's example glasses to within 1 to 4%, which is the acceptance gate for the property engine, and one example glass (Batch 45) is documented as a data provenance exception: its BBI-derived waste vector carries about 6% excess Al2O3 relative to the paper's formulation, and the normalized release metric is sensitive to alumina at high-zirconia compositions.

## Re-attribution under adjusted WAC

When a WAC limit is relaxed or tightened, campaigns must be re-blended without changing total waste throughput. The builder solves a quadratic program per campaign: minimize the squared distance between new and original layer volume fractions subject to preserving total campaign volume and satisfying every adjusted criterion linearly. On infeasibility, the worst-contributing layer per the violated criterion is dropped and the solve retries, up to three drops, after which the campaign is flagged rather than silently re-scoped. Per-layer WAC values are never regressed from published per-campaign weighted averages; the published blends are too uniform per criterion, and a regressor trained on them returns near-constant values that make the quadratic program unable to redistribute anything. Heel-layer WAC parameters are computed from BBI chemistry directly, with each criterion's own formula and volume basis (all-phase tank volume, not sludge-only).

## Stream thermodynamics core

The facility model's Stream carries mol/s per species, a phase per species, temperature in kelvin, and pressure in kilopascals. Enthalpy is computed as $H = \sum_i n_i\,[\Delta H_{f,i}(phase_i) + h_{sensible,i}(phase_i, T)]$. Formation enthalpies are referenced at 298.15 K; gases use NASA Shomate polynomials (seven or eight coefficient rows are both accepted, since standard tables append the 298.15 K heat of formation as an eighth column); condensed species use polynomial sensible heat. The distinction is not cosmetic: flash duties are computed as the enthalpy difference between the actual phase split and an all-vapor stream at the same temperature, both through the same component library, which is what makes latent heat close against the unit-level energy balance. Saturation pressure for water uses a correlation validated at steam-table anchor

points before entering any flash logic, with test tolerances set to the correlation's accuracy class rather than exact literature values.

Mass closure is enforced with hybrid absolute-relative tolerances: a per-species check passes if relative residual is below 1e-4 or absolute residual below 1e-9 mol/s. The hybrid form exists because trace isotopes at 1e-12 mol/s accumulate integration residue that a pure relative test would flag spuriously, while a pure absolute test would pass real leaks on major species. Every unit operation test asserts per-species conservation and an energy balance with the duty made explicit, and the whole train closes a reagent-inclusive element audit at its boundary: every external reagent (wash water, ammonia for SCR, quench make-up) joins the inlet ledger before products are compared, because an audit that forgets its own reagents reports hydrogen, oxygen, and nitrogen failures that are bookkeeping errors, not physics errors.

## The melter partition engine

The melter is the heart of the facility model and the algorithmically least conventional unit. Its input is the speciated campaign feed at the MFV draw rate; its outputs are a glass oxide stream, an off-gas stream, and the cold-crucible-equivalent joule duty. The algorithm runs in four stages. First, decomposition: every feed species is stripped to a per-element mass ledger (charge signs and hydration water are resolved into the ledger, isotope species pool onto their host element). Second, partition: for each volatile element, a scenario-supplied decontamination factor moves a fraction 1/DF of that element's ledger mass to the off gas, with the carrier molecule named per element (water family species, halogen acids, nitrogen oxides, carbon and sulfur oxides, selenium and cesium hydroxides); decontamination factor 1 means everything to gas, and large DF means near-total retention in glass. Third, synthesis: the off-gas products are built acids-first, halogen acids claim their hydrogen before water formation, and every synthesis step is capped at partner-atom availability from the ledger, so no synthesis can fabricate atoms; leftover nitrogen and oxygen sweep to N2 and O2, and retained halogens return to the glass as halide salts. Fourth, glass synthesis: the remaining ledger mass resynthesizes as glass oxides in a fixed order with the same partner-atom capping.

Two guards keep the engine honest. A residual-ledger gate asserts that unallocated mass stays near zero after synthesis; total mass closure alone can hide stranded elements, so the residual is a separate, explicit check. And melter particulate entrainment is applied after glass synthesis as a post-glass bleed: a fixed mass fraction (0.5%, the scenario's dust carryover parameter) of each glass oxide species moves to the off gas as condensed-phase dust at the same chemical formula, which conserves elements by construction and keeps dust loadings traceable into the downstream

particulate capture stages. Species that exist both as volatile gases and glass oxides (technetium heptoxide, selenium dioxide) collide as dictionary keys; the engine folds dust into the existing gas key rather than overwriting, a bug class caught by a real-campaign test.

Melter electrical power is computed by the duty approach: the joule duty equals the enthalpy rise from feed and glass-former inlets to glass plus off-gas outlets, divided by an electrical efficiency, rather than being modeled from electrode geometry. The feed-prep chain upstream is batch-discrete: the MFPV receives 3,800-gal slurry batches (vessel limit 5,100 gal), each batch is blended from the campaign recipe with glass-forming chemical powder and flush water (60 mL per 100 g of glass formers) added in-vessel, and the MFV draws the batches to the melter at a fixed volumetric rate (2.1 gpm nominal). Campaign runtime is computed from volume flow, not mass flow at a nominal density, because the blended density (feed plus powder plus flush, about 1.31 kg/L) differs from the nominal slurry density, and the difference is several percent of runtime. Batch counts are derived by dividing by the actual per-batch slurry draw, not the vessel capacity, with an explicit volume floor in the fixed-point iteration, two guards added after the iteration oscillated on a powder-heavy heel campaign and silently dropped most of the glass-former dose.

## The off-gas train unit by unit

Downstream of the melter, each unit applies capture factors and thermodynamics on the same Stream basis. The submerged bed scrubber quenches gas to 100 °F held by a cooling sink, with the outlet saturated at bath temperature and reduced pressure (a 35 in. water column loss). Capture is per-species decontamination factors into the scrub liquor; captured species land in the liquor as aqueous-phase species of the same formula, which keeps element conservation exact and defers dissociation and pH to the downstream effluent chemistry. No reagents are dosed, and the 90 gpm internal recirculation is a mixing attribute, not a stream crossing the material balance. The bath set-point is held by a saturation solve at the outlet state (gas water content from the saturation pressure at bath temperature and outlet pressure, remainder to the liquor), never by a fictitious quench-water feed.

The wet electrostatic precipitator and the mist eliminator share a wet-stage engine: per-species capture, wash-water addition (subcooled supply at a specified temperature), re-saturation control (the gas tops up from the wash pool first), and a unit enthalpy balance. The mist eliminator is a droplet separator, capturing only liquid and aqueous droplet phases while gases pass at unit transmission. A reheater adds a fixed dew-point margin (+36 °F), its duty computed from the stream enthalpy difference. The HEPA filter books all particulate capture (decontamination factor

10,000) as a media loading ledger rather than deleting mass. Fan shaft work enters the gas as sensible heat, with outlet temperature Newton-solved from the enthalpy balance and the dimensional identity stated in the code: kilopascals times cubic meters per second is kilowatts.

The packed beds (activated carbon for mercury, silver mordenite for iodine) run through one shared bed engine with per-species capture factors, loading ledgers in kilograms per campaign, and pressure drops. The hot block couples five units in one solve: the regenerative heat exchanger's cold leg, a trim heater, the selective catalytic oxidizer (650 °F set-point), the selective catalytic reducer, and the hot leg returning to fan 2. Because the hot side is downstream of the cold side, the block is solved as one fixed-point iteration anchored at the reactor inlet set-point (the trim heater tops up whatever the exchanger delivers); solved sequentially, an uninitialized hot-inlet temperature feeds the cold duty and the loop diverges. The reactors are modeled isothermal at their set-points with reaction enthalpy exported as an explicit sink duty that is returned to the gas stream at the outlet. SCR stoichiometry injects ammonia per the NOx inventory with a slip fraction, and the slip is tracked to the stack as an emission rather than absorbed.

## Effluent chemistry and transfer scheduling

Condensate from the scrubber, precipitator, and mist eliminator accumulates in the condensate vessel. Transfers are scheduled at vessel high level (6,000 gal) plus a campaign-end partial transfer, and the ledger reports a campaign-average effluent composition. Chemistry is computed at equilibrium: strong-acid concentration from the strong-acid anion inventory, pH from a charge balance over the acid-base species present (nitrate, nitrite, sulfate, phosphate, oxalate, chloride, fluoride, carbonate, hydroxide), with dissociation equilibria for the weak acids. The model reproduces the acidic condensate expected on heel campaigns, measured campaign-mean pH in the 0.9 to 2.5 range across the mission on the reference scenario, and the wash sweep below shows how that number moves with retrieval practice.

## The discrete operations layer (Beta)

The physical model above is continuous in streams and batch-discrete in vessels. The discrete layer adds the calendar: an operator-procedure catalog read from a YAML data file (batch receipt, glass-former dosing, in-vessel blending, transfer to MFV, melter pour, canister handling, campaigns turnaround), a scheduler that folds the procedures serially over the physical model's own batch volumes, and a coupling module that consumes MFV batch arrivals. Two design rules keep it honest. Mass invariance: the layer recomputes nothing about mass or composition, it consumes the physical chain's batch volumes and species totals and gates itself on drawn-species

totals matching the continuous chain to better than 1e-6 relative, so it answers when, not how much. Capacity enforcement: when a pour schedule exceeds the canister-handling ceiling, the scheduler flags it while aspirational, and when enforcement is enabled it folds cumulative output against absolute time (an exact cumulative bound rather than per-batch spacing), which emits the enforced idle time as a visible catch-up event instead of hiding it inside per-campaign averages.

Discreet modeling is mentioned in this paper, but is a topic of future model development and sensitivity analysis.

### Wash efficiency and heel re-synthesis (Tanks Farms NE, NW, and SW)

The wash-efficiency control is the newest coupling between the applications (campaign development and HLW facility modeling). For a given scenario and a wash efficiency value (the fraction of soluble heel species decanted to LAW during retrieval, NE/NW/SW quadrant practice), the overlay re-synthesizes the heel layer records by calling the campaign builder's own heel-composition flowsheet with the value substituted, producing heel vectors with the same species conventions and data shapes as the shipped files. At the default value (0.9) the shipped vectors are used byte-identically, which keeps the published basis the default path. The campaign plan itself, the recipe and its fractions from Section 6.2, is deliberately held fixed; wash changes heel chemistry only, so every mission-level delta in the sweep is attributable to wash alone rather than to a re-planned schedule. The overlay is cached per scenario and efficiency value, and any trends sweep computed at a non-default efficiency bypasses the persistent cache to prevent a swept basis from overwriting the published one.

### Stream monitoring and master export

The monitoring chain exposes three computed streams per campaign: the blended slurry feed (pre-dosing), the MFPV outlet (campaign mean over all batches, including glass-former powder and flush), and the MFV outlet (the steady draw the melter sees). The campaign-mean outlet is computed as total moles per species divided by total runtime, and its consistency with the MFV draw is a test gate: the two compositions agree to within 0.1% molar, and the mass chain, feed plus powder plus flush against the outlet, closes to 100.00% on the reference campaign. The trends engine adds per-campaign series for both streams (mass flows, top-species mass fractions, acid-analyte campaign totals, decant context), and the interactive process flow diagram renders hover composition on the feed-prep vessels from the same streams. The master export writes one Excel workbook per campaign with fourteen sheets: a provenance readme, stage states, each major stream (feed, MFPV outlet, MFV outlet, melter off gas, glass, effluent, stack), media loadings, a per-element closure audit, the batch ledger, per-unit energy, and the heel wash/decant

ledger. Every species row carries both the rate basis (mol/s, g/s) and the campaign-integrated mass, so a reviewer can spot-check a mass balance from the workbook alone.

## Verification strategy

The test suite is the verification strategy: 147 tests run on every change, in seconds, and nothing ships until the full battery passes. The layers are: unit tests per unit operation (per-species mass closure, energy closure with explicit duty, physical anchors such as water boiling point and steam-table saturation checks); chain tests that drive a real campaign through the whole train and audit elements at the train boundary (reagent-inclusive); regression tests over the campaign trends that catch silent changes in mission totals; and application tests that execute the Streamlit app headlessly and assert no exception on any tab, which is how heel-only-campaign crashes and late-campaign data-path bugs were caught that single-campaign tests missed. Property-model correctness has its own gate: the EWG2.5 engine must reproduce the published example-glass property table within 1 to 4%, and a campaign-band check asserts every campaign inside the 450 to 550 kgal volume band. When a verification number disagrees with a hand-computed anchor, the dimensional identity is written into the helper function and pinned with a regression test, several of the guards described above exist only because a plausible-looking wrong number once passed every static check.

# Results

## Mission-level outputs for two published scenarios

Table 1 summarizes full-mission results for the two reference scenarios, computed by running every campaign of each scenario through the complete facility chain. Scenario tags encode supernatant management, target waste loading, and solids loading; 135.120.150 is the heavier condition (higher target waste loading), and its mission is larger and longer in every column. The scenarios differ in both the final liquid specific gravity and expected suspended solids. The 135.120.150 condition is closer to an expected nominal condition given little washing and system optimization. With greater carrier fluid density there is less risk of settled solids during campaign preparation. On the other hand, 135.105.200 shows performance of an optimized stream in which the supernate density is minimized, while increasing the solids loading (high level waste sludge) in the feed.

*Table 1. Full-mission results from the coupled model (default wash efficiency 0.9).*

| Scenario | Campaigns | Glass (kt) | Canisters | Effluent (kt) | Runtime (d) |
|---|---|---|---|---|---|
| 135.105.200 | 59 | 45.0 | 14,526 | 92.1 | 9,044 |
| 135.120.150 | 76 | 53.5 | 17,287 | 119.8 | 11,524 |

Runtime here is the melter-processing duration: the sum of campaign runtimes from the volume-flow computation described in the Methodology, and it excludes window-dressing calendar effects such as facility commissioning or inter-scenario transitions. The canister counts use the 3,097 kg planning mass per canister. Campaign counts are read live from the campaign builder.

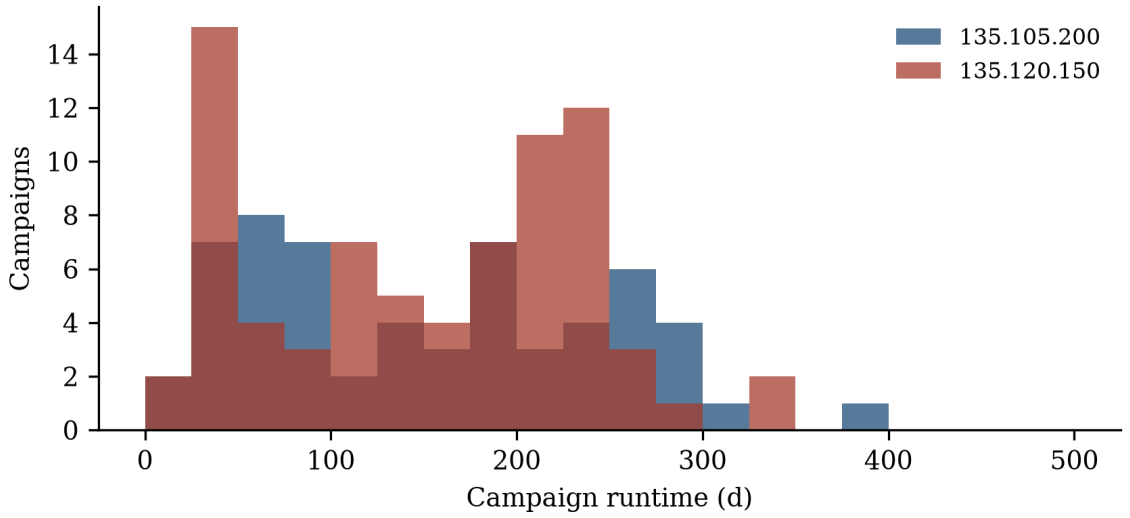



*Figure 1. Campaign runtime distribution for both reference scenarios. The longer tail on 135.120.150 reflects heel-heavy late campaigns and the larger per-campaign waste volume of the higher waste-loading condition.*

## Wash-efficiency sensitivity across the mission

The wash sweep re-runs every campaign of a scenario at four wash efficiencies, with the retrieval plan held fixed and only the heel records re-synthesized. Table 2 gives mission rows; Figure 2 plots glass and runtime against efficiency.

*Table 2. Mission-level wash-efficiency sweep (campaign plan fixed; heel records re-synthesized per value).*

| Scenario | Wash η | Glass (kt) | Canisters | LAW decant (kt) | Heel slurry (kt) | Feed NO3 (kt) | Effluent pH | Runtime (d) |
|---|---|---|---|---|---|---|---|---|
| 135.105.200 | 0.5 | 86.2 | 27,818 | 77.4 | 118.2 | 17.6 | 0.63 | 16,033 |
| 135.105.200 | 0.7 | 58.1 | 18,757 | 100.9 | 86.6 | 9.7 | 0.78 | 12,097 |
| 135.105.200 | 0.9 | 45.0 | 14,526 | 132.7 | 46.7 | 2.4 | 1.29 | 9,044 |
| 135.105.200 | 1.0 | 44.8 | 14,480 | 148.9 | 26.4 | 0.5 | 1.83 | 7,902 |
| 135.120.150 | 0.5 | 107.0 | 34,553 | 56.5 | 139.1 | 22.8 | 0.58 | 19,890 |
| 135.120.150 | 0.7 | 81.3 | 26,250 | 79.2 | 108.3 | 15.8 | 0.64 | 16,130 |
| 135.120.150 | 0.9 | 53.5 | 17,287 | 117.1 | 62.2 | 5.8 | 0.90 | 11,524 |
| 135.120.150 | 1.0 | 53.3 | 17,219 | 140.1 | 35.2 | 2.7 | 1.05 | 9,958 |

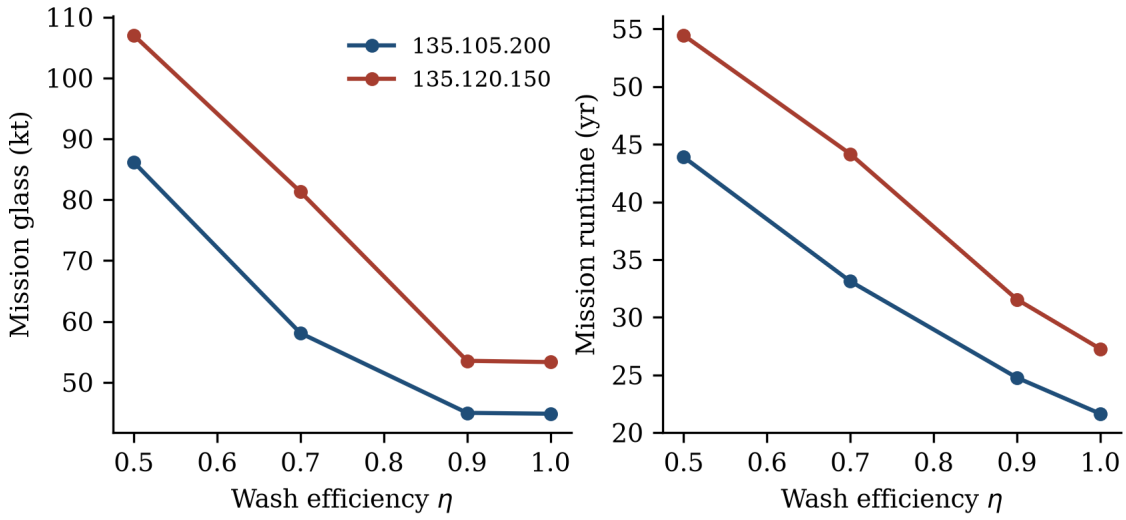



*Figure 2. Mission glass (left) and runtime (right) versus heel wash efficiency for both reference scenarios. The retrieval campaign plan is fixed in every run; only the heel chemistry changes.*

The shape of these curves tells a story of high sensitivity on retrieval and blending performed to retrieve the NW, SW, and NE farms. Mission glass is nearly flat between efficiency 0.9 and 1.0 on both scenarios (45.0 to 44.8 kt on 135.105.200; 53.5 to 53.3 kt on 135.120.150) but rises steeply below 0.9: at 0.5, glass grows by 91% on 135.105.200 (45.0 to 86.2 kt) and 100% on 135.120.150 (53.5 to 107.0 kt). Runtime scales similarly, from 9,044 to 16,033 days (24.8 to 43.9 years at the modeled processing rate) on 135.105.200. The mechanism for this stark contrast in performance is sodium partitioning: undecanted sodium nitrate and nitrite ride the heel into the HLW feed, consume glass-former capacity as soda, and lower effective waste loading campaign by campaign. The effluent story is the mirror image: at efficiency 0.5, decant to LAW drops from 133 to 77 kt on 135.105.200 while mission effluent rises from 92 to 151 kt and campaign-mean pH falls toward 0.6, because the acid anion load that would have left with the decant instead volatilizes in the melter and exits through the wet train. Nitrate in the melter feed rises from 2.4 to 17.6 kt across the same sweep, which propagates to the SCR ammonia demand and the stack NOx numbers below.

Two practical readings follow. First, the marginal value of washing improvement is concentrated between efficiencies of about 0.5 and 0.9; pushing from 0.9 to 1.0 buys almost nothing in glass and canisters, because the remaining heel solubles at 0.9 are already small. Second, the mission is remarkably insensitive to wash quality in runtime terms once a floor of roughly 0.9 is achieved, which suggests the retrieval organization can trade wash thoroughness against retrieval rate below that floor only at a steep HLW-scope price.

## Melter-feed monitoring chain

On the reference campaign (135.105.200, campaign 40, a heel-heavy campaign with five heel layers in the recipe), the chain closes exactly: feed slurry mass over the campaign (2,612,524 kg) plus glass-former powder and flush (976,800 kg) equals the campaign-mean MFPV outlet mass (3,589,324 kg), a 100.00% closure, and the MFPV and MFV outlet compositions agree to a maximum relative difference of 9.5e-4 molar. Figure 3 shows the closure graphically.

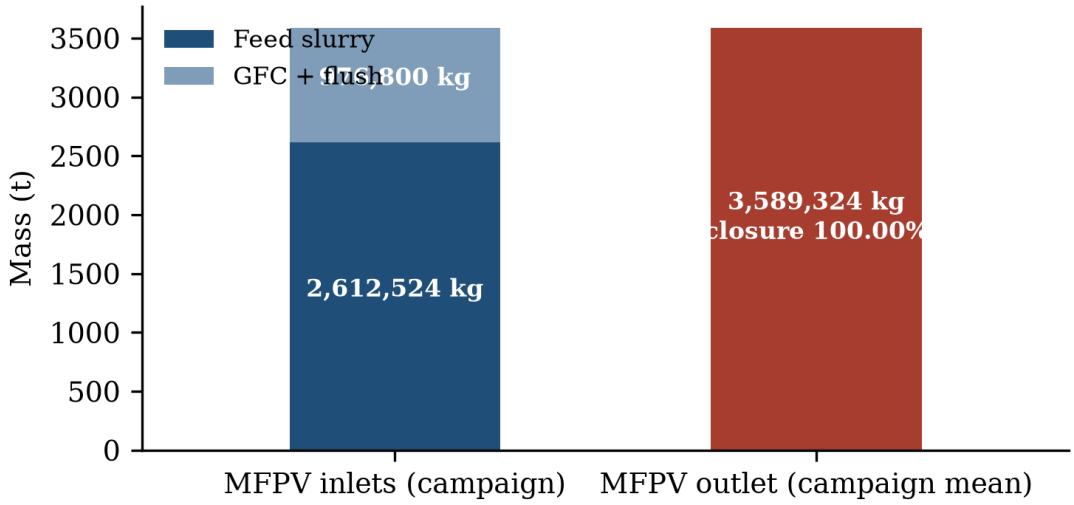



*Figure 3. Melter-feed chain mass closure on campaign 40 of 135.105.200. The MFPV inlet bars (feed slurry, then glass-former powder plus flush water) sum to the campaign-mean outlet mass.*

The monitoring chain earns its keep on composition detail. On this campaign the MFV draw carries the sodium and aluminum signature of the heel oxides plus the dosed boron and lithium, and the feed-versus-dosed comparison makes the dosing delta visible species by species. The same streams feed the per-campaign trend series, where the top six species by mass fraction are tracked across the mission; the acid anions (nitrate, sulfate, oxalate) appear in the feed series on heel campaigns and vanish on App-D-only campaigns, which is the campaign-level signature of wash behavior in the upstream data.

## Glass composition and effluent chemistry over the mission

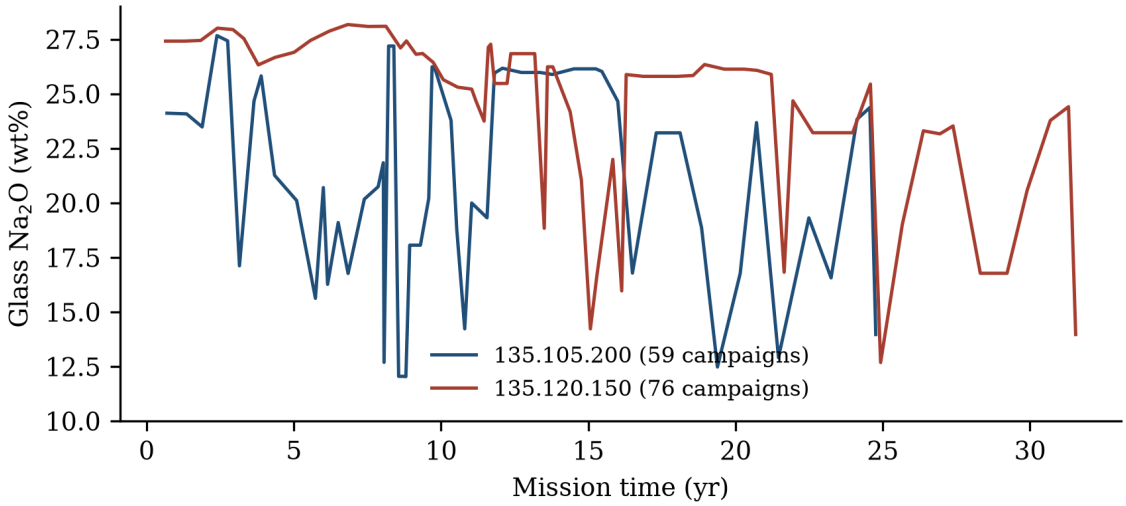



*Figure 4. Glass sodium oxide mass fraction by campaign across the mission for both scenarios.*

Glass soda (Na2O) spans 12.0 to 27.7 wt% across campaigns on 135.105.200. The high-soda campaigns are the heel-heavy ones late in the mission, where undecanted sodium salts concentrate in the feed; the WAC-validity bound (27% soda maximum in the composition model) is the binding constraint there, and campaigns that would exceed it are held at the bound with correspondingly lower waste loading. The low-soda campaigns early in the mission are supernatant-rich App-D layers where aluminum and silicon dominate.

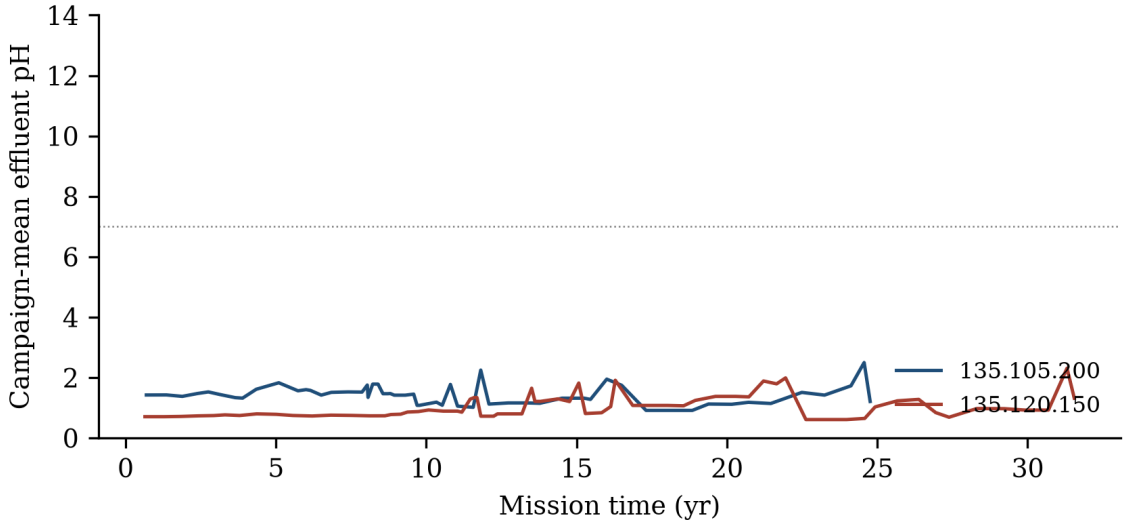



*Figure 5. Campaign-mean effluent pH across the mission. Both scenarios run acidic condensate throughout; the deep dips are heel-heavy campaigns.*

Effluent pH runs 0.92 to 2.5 across the mission on 135.105.200 (0.58 to 1.05 across the sweep's mission rows on 135.120.150 at default wash). The condensate is a strong-acid solution: nitrate, nitrite, sulfate, and chloride anions that volatilized from the feed re-dissolve in the wet-train liquor.

## Off-gas performance

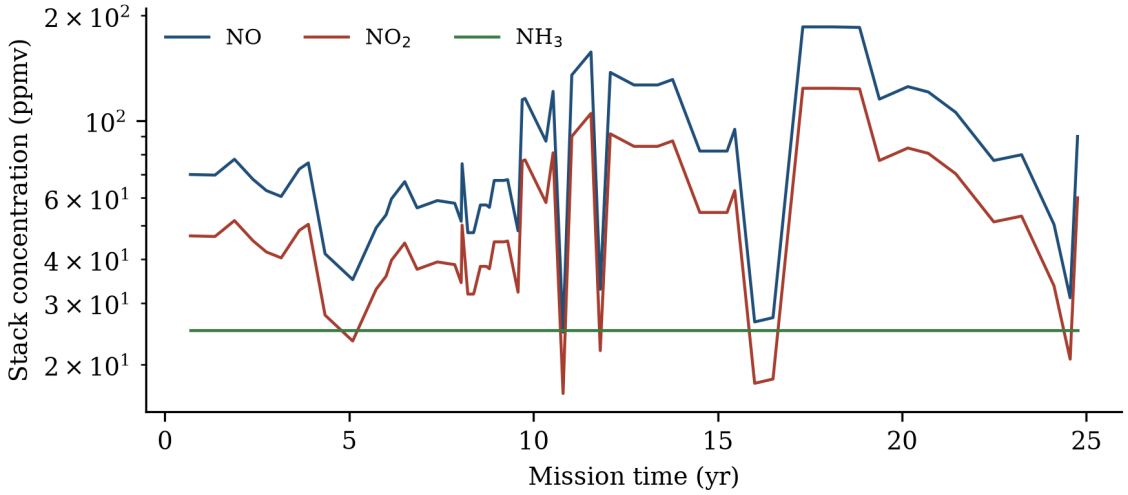



*Figure 6. Stack concentrations of NO, NO2, and NH3 by campaign on 135.105.200 (log scale).*

Stack NO averages 80.8 ppmv (53.9 for NO2) with ammonia slip averaging 25.0 ppmv across the mission and a fixed removal efficiency of 95% assumed for the SCR catalyst. The chain-derived expectation is that stack NOx equals the melter off-gas NOx times the product of stage transmissions (scrubber 1/2, SCR residual 5%); the mission-level numbers sit inside that envelope, with campaign variation tracking feed nitrate, heel campaigns run higher. Ammonia slip tracks SCR injection stoichiometry against the per-campaign NOx inventory.

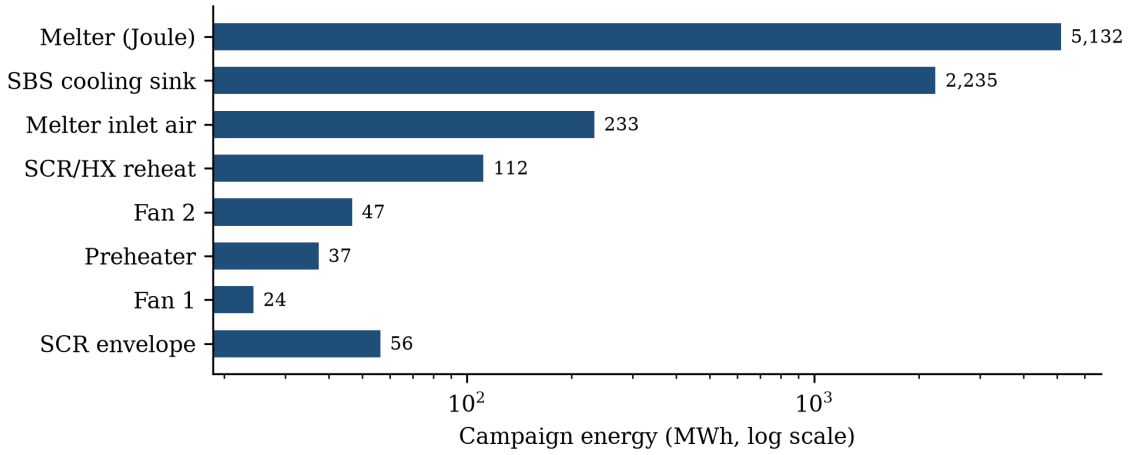



*Figure 7. Campaign energy by consumer on campaign 40 of 135.105.200 (log scale). The joule duty dominates; the scrubber cooling sink is the second-largest term.*

Energy on the reference campaign is dominated by the melter joule duty (5,132 MWh over a 234-day campaign, an average of 913 kW) with the scrubber cooling sink second (2,235 MWh). Mission-wide on 135.105.200, the melter consumes 166,964 MWh and the net facility power draw is 185,416 MWh over 9,044 days. The regenerative heat exchanger recovers the reactor-block enthalpy and returns it upstream, which is why the trim heater duty (112 MWh per campaign) is two orders of magnitude below the exchanger's service.

## Verification evidence

*Table 3. Verification evidence: whole-mission element closure on 135.105.200 (59 campaigns) and test-suite composition.*

| Check | Result | Gate |
|---|---|---|
| Elements closed < 1% relative | 37 of 38 | whole-train audit |
| Worst closure (As, trace 0.2 mg/s) | 21.2% rel = 1 µg/s abs | hybrid tolerance pass |
| Worst closure (major element) | H, 0.92% (c8) | < 1% pass |
| Melter residual ledger | ≈ 0 on all 59 campaigns | explicit gate |
| Feed + GFC + flush = MFPV out | 100.00% (c40) | chain closure |
| MFPV out = MFV out (composition) | < 1e-3 molar (c40) | stream identity |
| Total campaign mass closure | exact, all 59 | ledger identity |
| Automated test suite | 147 tests, all pass | every change |
| App headless execution | no exception, all tabs | AppTest battery |

The whole-mission element audit (Table 3) closes every tracked element to better than 1% relative except one trace case: arsenic on campaign 8 shows a 21.2% relative residual that is 1 microgram per second in absolute terms, and the hybrid tolerance (relative 1e-4 OR absolute 1e-9) correctly passes it. Hydrogen shows residuals up to 0.9% on a few campaigns; hydrogen is the hardest element to close because it crosses every phase boundary (feed water, hydrate dehydration, acids-first off-gas synthesis, flush water) and its ledger absorbs the roundoff of every stage. The residual-ledger gate in the melter engine stays near zero on all 59 campaigns, and total mass closes exactly on every campaign, which is the property that makes the per-element residuals interpretable as bookkeeping roundoff rather than leaks.

## The process flow diagram as a live model view

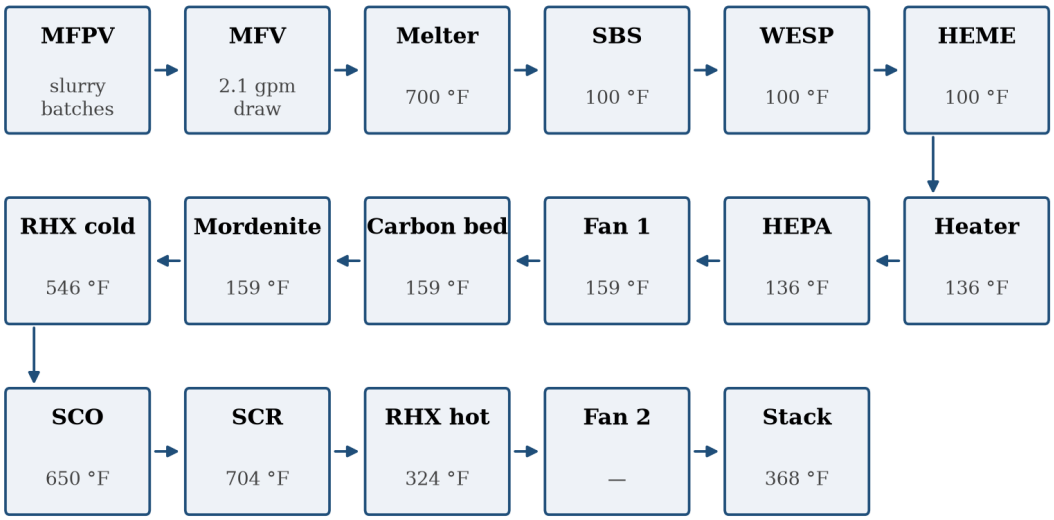



*Figure 8. Modeled process topology with campaign-40 stage temperatures on 135.105.200. The interactive application version renders hover compositions and flows from live model state.*

The topology shows the modeled temperature profile at a glance: melter outlet at 700 °F, quench to 100 °F through the wet train, reheat to 136 °F for particulate filtration, fan work recovering to 159 °F, ambient-temperature beds, and the reactor block at its 650 °F set-point with the 704 °F SCR outlet discharging through the regenerative exchanger (324 °F) and fan 2 to the stack at 368 °F. Each of these numbers is a model output computed from the enthalpy balances described in the Methodology, not a transcribed PFD value, and the interactive version in the application recomputes them for whichever campaign the engineer selects.

# Discussion

## Interpretation

The modeling chain earns its complexity in exactly one place: the coupling. A standalone facility model fed a generic waste would reproduce the plant physics but say nothing about the mission; a standalone campaign planner would optimize blends against acceptance criteria with no visibility into what those blends do downstream. The coupled pair turns retrieval planning questions (how well must we wash? which layers should ride together?) into plant-output questions (glass tonnage, canister count, stack compliance, effluent acidity) with one auditable data path. The wash sweep is the clearest demonstration: the difference between 45 and 86 kt of glass is not a facility-model effect at all, it is a retrieval-practice effect that only exists because the heel chemistry reaches the melter model unchanged in kind but changed in proportion.

The verification posture also shapes interpretation. Every headline number in this paper is reproducible from cached campaign outputs and re-derivable from the test suite's anchors. Where the model and a hand-computed anchor disagreed during development, the cause was invariably a dimensional or basis convention (a rate treated as a total, an oxide yield applied twice, a unit conversion absorbed silently), and each such discovery became a test and a guard. The numbers reported here carry the confidence of that process, and the guards now in place mean the same class of error cannot silently recur.

## Limitations and model scope

The model is a steady-per-batch process model, not a transient one. Within a batch, the melter sees a constant draw at the MFV rate; true feed variability within a batch (the last, solids-enriched liter of a vessel, for instance) is outside the model's resolution. The effluent chemistry is equilibrium-based at transfer time; kinetic effects (crystal precipitation, slow redox) are not modeled, and the pH values for highly concentrated acid condensates inherit the accuracy limits of the correlation set. The melter partition factors are scenario-supplied engineering values, not first-principles calculations; they are the right tool for sensitivity work but should not be read as predictions of absolute emission rates. Stack concentrations inherit that caveat directly.

The campaign plan for SE quadrant is taken from the published RPP-RPT-64878 schedule rather than re-optimized jointly with the facility model. Future work will focus on targeting blending operations in tanks farms, using public BBI data and previous work in RPP-RPT-64878. A joint optimization (choose the plan that minimizes glass or cost subject to plant constraints) is the obvious next step and is now computable, since a full mission evaluates in about 76 seconds, but

it is not what the current results report. The wash sweep deliberately holds the plan fixed so deltas are attributable to wash alone; a re-planned schedule under poor wash would shift some of the runtime growth shown here.

## Data provenance and basis caveats

Three basis conventions deserve explicit statement for anyone comparing numbers across documents. First, the App-D hydrate convention: the published oxide vectors treat hydrated analytes (aluminum hydroxide and its relatives) as elemental mass converted to oxide, which is the convention the glass targets were optimized under; the facility model's speciated feed keeps hydrates molecular, and the two conventions differ by several percent of aluminum and silicon on hydrate-heavy layers. The difference is a basis-convention artifact, documented and adjudicated with the SME, not a partition error. Second, canister mass: 3,097 kg per canister is the App-D planning value (published glass tonnage divided by published canister count), not a nominal 2.5 or 3.0 t. Third, glass composition sums to less than one by the volatile fraction (water and halogens leave the melt); the closure term is physical and is retained rather than renormalized away.

The Batch 45 example-glass exception (normalized release 1.37 versus the published 1.61) traces to the BBI-derived waste vector carrying about 6% excess Al2O3 against the paper's formulation; it is documented as a data caveat with the mechanism identified (alumina sensitivity at high zirconia) rather than tuned away. One example passing while its siblings reproduce exactly is the signature of a data provenance difference, not a model defect, and treating it as a model bug would have corrupted the property engine to chase a data artifact.

## Computational performance

A full campaign runs in 1 to 2 seconds and a 59-campaign mission in about 76 seconds on a desktop workstation (Apple silicon, single process). The dominant cost is the EWG2.5 property evaluation under the multistart SLSQP solve; earlier profiling showed the property engine re-parsing static coefficient tables per constraint evaluation, and caching the parsed matrices plus vectorizing the quadratic forms cut a campaign evaluation from 14.2 to 1.14 seconds (12.5x), which is what makes whole-mission sweeps and the interactive application practical. The trends layer caches per-scenario campaign sweeps as parquet files and validates row counts against the live campaign source on load, so a stale cache presents as a rebuild, not as wrong numbers.

## Application experience and review workflow

The applications are built for review, not just for computation. The Streamlit interface puts the process flow diagram, live stream tables, and the operations schedule side by side so a reviewer can trace any number on the diagram to a table and to the master workbook. The export workbook carries provenance (scenario, campaign, wash basis, runtime, campaign source) on its readme sheet, so a workbook pulled from the application is self-identifying months later. The discrete operations schedule renders as a per-batch timeline with procedure labels, and the enforced pour-ceiling case makes capacity limits visible as explicit catch-up events rather than silently stretching the calendar.

## Conclusions

Two applications, a campaign builder and a facility process model, now cover the Hanford HLW mission continuously from tank inventory chemistry to stack and schedule. The architecture choices that made this work are a single speciated-stream thermodynamic core shared by every unit operation; an element-ledger melter partition engine with partner-atom-capped synthesis and a residual-ledger gate; a campaign bridge that preserves species identity from the planner's analyte inventory into the plant model; and a verification battery that runs on every change and closes elements at the train boundary with reagents included.

The quantitative findings the team can now stand behind are mission-level and decision-relevant. Wash efficiency is the dominant single lever on HLW scope available to the retrieval organization: mission glass rises 91 to 100% and runtime 77 to 79% when wash efficiency falls from 0.9 to 0.5 across the two reference scenarios, while improvement beyond 0.9 buys almost nothing. The melter-feed monitoring chain closes mass exactly and exposes composition at the three points that matter for feed control. The off-gas train holds stack NOx inside the chain-derived envelope with ammonia slip tracked as an emission, and effluent acidity is now a predicted, campaign-resolved quantity at the effluent interface rather than an assumed one.

The development path forward runs through the coupling itself: joint plan-and-plant optimization is now computable at 76 seconds per mission; effluent transfer chemistry return interface loading deserve the same speciated treatment the off-gas train received; and the partition-factor library will grow in value as each scenario submission refines it.

## References

RPP-RPT-64878, Tank Waste Operations Modeling (TWOM): Development of Overall Integrated Flow sheet and Mission Analysis for Direct HLW Feed, Washington River Protection Solutions, Richland, WA. (Appendix D: layer-by-layer feed vectors and analysis; Appendix E: campaign blend sheets; Section 6.2: campaign generation methodology.)

PNNL-35884, Enhanced Waste Glass (EWG) Model Version 2.5 for HLW Glass Property and Composition Estimates, Pacific Northwest National Laboratory, Richland, WA. (Property models: product consistency, viscosity, electrical conductivity, sulfate, phosphate; Appendix C: coefficient covariance matrices.)

PHOENIX Best-Basis Inventory data service, Pacific Northwest National Laboratory, Hanford Site tank waste inventory retrieval (current snapshot and all-tank records).

HLW System Model source repository, hlw-facility-mission-model, discrete-model/main branches (unit-operation modules, melter partition engine, discrete operations layer, verification suite).

EWG campaign builder source repository, HWG_mission_planning (campaign generation, WAC re-attribution, glass formulation, scenario synthesis).

NASA Glenn Research Center, thermodynamic property coefficients (Shomate polynomials) for ideal-gas species, NASA/TP-2002-211556.

U.S. Department of Energy, Office of River Protection, Waste Treatment Plant and tank farm mission planning basis documents for the Hanford Tank Waste Operations mission.